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# A Capable, Fully Tensor-Decomposable Vision-Language-Action Policy

Exact rational conversion, protocol-aligned control, and weight-derived  
analysis

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## Abstract

Tensor decomposition factors a large table of interactions into a small set of low-rank components that can be ranked and inspected. Applying it directly to a conventional robot policy is difficult because softmax, ordinary activations, and normalization interrupt the required algebraic form. We introduce  $\chi$ -VLA, a compact vision-language-action policy whose learned map uses only bilinear, linear, and rational operations. Its checkpoint therefore defines an explicit rational tensor program, making interactions among image, instruction, state, and action coordinates accessible from the weights. A mechanical audit finds no hidden incompatible operation, and local reconstructions agree near the float32 precision floor. On all ten LIBERO-Object tasks, the convolutional 19.2M-parameter instantiation reaches  $93.7\% \pm 0.8\%$  under the published three-seed protocol. Averaging its three fixed checkpoints reaches 96.2% at three times the inference cost. On the three released convolutional checkpoint files associated with those historical aggregates, independent reconstruction covers all 4,608 joint-attention head-input cases with worst relative error  $6.36 \times 10^{-16}$ . In a separate three-checkpoint ViT instantiation, we derive an exact weight-based decomposition of softmax-free attention and reconstruct every attention module with worst relative error  $2.56 \times 10^{-7}$ . All 36 primary block-head tests favor its rank-128 subspace over equal-rank random controls, with median fidelity ratio 2.44. The evidence establishes strong specialist capability and exact layerwise access to trained attention coefficients. It does not establish broad robotic generalization, robust instruction grounding, or an exact end-to-end decomposition of the full policy.

## 1 Introduction

Modern robot policies can work well while remaining difficult to inspect. A checkpoint contains millions of learned numbers, but it does not directly reveal which combinations of image regions, instruction tokens, and robot state produce an action. Analysts therefore usually collect activations on a dataset and fit a separate probe. That can identify correlations, but it leaves open whether the same structure is present in the weights and whether it causes behavior.

Tensor decomposition offers a different analysis surface. A tensor is a multidimensional table, and a decomposition rewrites that table as a sum of simpler low-rank factors, much as a matrix factorization finds dominant directions in a matrix. If a network is built from additions, multiplications, and ratios of polynomials, its weights define an explicit table of input interactions. Decomposing that table can rank the combinations the model is constructed to use. By *fully tensor-decomposable*, we mean that

every learned block preserves this algebraic form. The aim is not merely compression. It is a direct path from parameters to candidate mechanisms that can be tested by intervention [3, 4].

Robot control makes this goal useful and difficult. Visual shortcuts can produce high benchmark success while ignoring the instruction, and small action errors compound in closed loop. Standard softmax, nonlinear activations, and square-root normalization break the explicit polynomial form. Continuous actions also cannot rely on the external categorical sampling used by a language model. A practical design must recover these functions without losing precise control.

We study the narrower but testable question:

*Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis?*

We test capability and structural analysis in two fully rational encoder instantiations. The convolutional checkpoints establish closed-loop capability. Separate ViT checkpoints support the exact attention audit.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate fully rational 19.2M convolutional and 20.1M ViT policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches  $93.7\% \pm 0.8\%$  over three seeds, and a three-checkpoint ensemble reaches 96.2%.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

## 2 A rational tensor robot policy

### 2.1 Tensor-compatible primitives

Each primitive below replaces a familiar transformer component while keeping an explicit coefficient description. Linear maps contribute first-order terms. Multiplying learned linear projections creates higher-order interactions whose coefficients remain functions of the weights, rather than an unstructured nonlinear black box.

**Homogeneous coordinate.** We write  $\bar{x} = [1; x]$ . The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

**Bilinear feed-forward block.** For  $L, R \in \mathbb{R}^{r \times (d+1)}$  and  $D \in \mathbb{R}^{d \times r}$ ,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left( \sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is an order-three table indexed by output coordinate and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization, without materializing every table entry.

**Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

Here  $M$  is a fixed causal mask,  $C$  is fixed row scaling, and  $s$  is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations. Unlike softmax attention, every multiplicative interaction can be expanded into coefficients determined by the trained weights.

**Rational per-instance normalization.** RMS normalization contains an input-dependent square root. Let  $m(x) = d^{-1} \sum_{j=1}^d x_j^2 + \epsilon$ , and let  $s_0 > 0$  be the frozen running mean-square scale. We deploy a fixed rational approximation  $r(v) = P(v)/Q(v)$  to  $v^{-1/2}$  over  $v \in [0.1, 10]$ :

$$v = \frac{m(x)}{s_0}, \quad \text{RNorm}(x) = \frac{x}{\sqrt{s_0}} r(v). \quad (3)$$

Here *rational* means a ratio of two polynomials. Projective coordinates carry each activation as a numerator-denominator pair  $(N, D)$  representing  $N/D$ . Linear, bilinear, residual, and normalization operations update this pair. The model therefore keeps input-specific rescaling while remaining exactly representable as a rational tensor network.

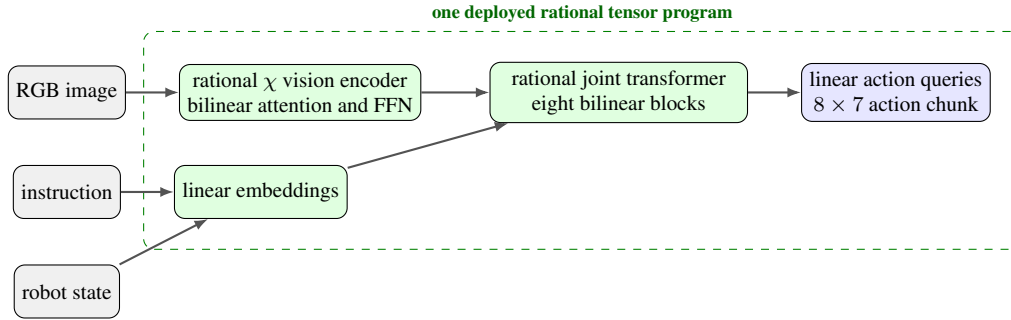


Figure 1: **Deployed  $\chi$ -VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

## 2.2 Architecture and training

The policy follows the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. It uses a  $64 \times 64$  agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven continuous action dimensions. The complete model has 20,137,352 learned parameters.

Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a  $180^\circ$  camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

## 2.3 Mechanical certificate on the 20.1M ViT checkpoint

Architecture definitions can differ from deployed code. We therefore audit the real 20.1M ViT 189/200 checkpoint at runtime and reconstruct representative operations from saved weights.

### Mechanical decomposability audit of the trained 189/200 checkpoint

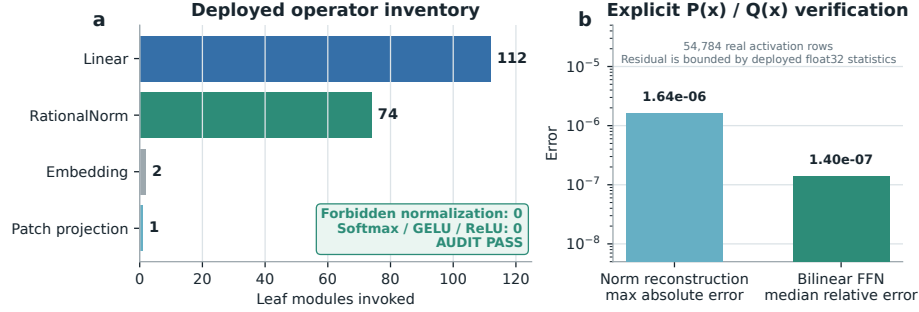


Figure 2: **Mechanical audit of the 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

98 The normalization reconstruction has maximum absolute error  $1.64 \times 10^{-6}$ . A bilinear FFN re-  
 99 construction has median relative error  $1.40 \times 10^{-7}$ . These residuals are bounded by the deployed  
 100 module’s float32 statistic. The audit certifies operator membership and local reconstruction. It does  
 101 not materialize one compact polynomial for the full eight-layer transformer.

## 3 Protocol-aligned closed-loop capability

### 3.1 Evaluation protocol

104 We evaluate all ten LIBERO-Object tasks with:

- 105 • 50 canonical trials per task
- 106 • a 280-step cap
- 107 • three independently trained seeds per architecture
- 108 • 500 rollouts per seed and 1,500 rollouts per architecture

109 Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not  
 110 rerun in our codebase, so differences may still include implementation and training choices.

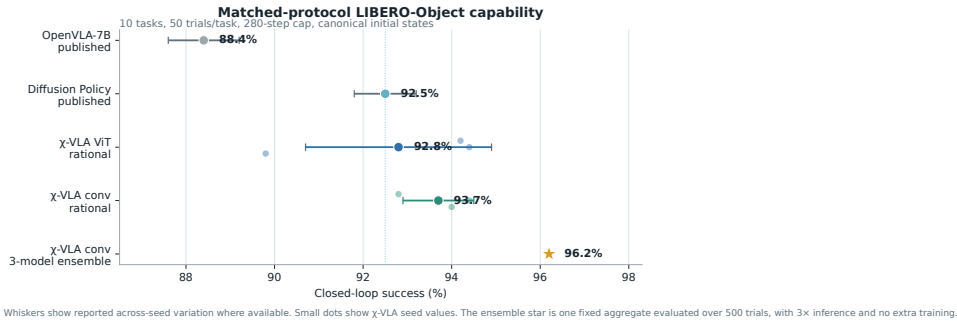


Figure 3: **Protocol-aligned LIBERO-Object capability.** Small dots are  $\chi$ -VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

## 111 3.2 Results

| Method                     | Parameters              | Training seeds    | Success                            |
|----------------------------|-------------------------|-------------------|------------------------------------|
| $\chi$ -VLA, rational conv | 19.2M                   | 3                 | <b>93.7% <math>\pm</math> 0.8%</b> |
| $\chi$ -VLA, conv ensemble | $3 \times 19.2\text{M}$ | $3 \rightarrow 1$ | <b>96.2%</b>                       |
| OpenVLA-7B [1]             | 7B                      | 3                 | 88.4% $\pm$ 0.8%                   |
| Diffusion Policy [1]       | not matched             | 3                 | 92.5% $\pm$ 0.7%                   |

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

112 The three historical convolutional aggregate records report 94.0%, 94.4%, and 92.8%. Their narrow  
113 spread is descriptive bounded evidence across independently trained checkpoints.

114 Separately, the paired ensemble evaluation re-evaluates the three members at 93.6%, 95.0%, and  
115 93.2%, for a 93.9% mean. Elementwise prediction averaging reaches 96.2%. This is 2.3 points  
116 above that paired mean and 1.2 points above its strongest member. The gain is largest on tomato  
117 sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one  
118 19.2M-parameter policy.

## 119 3.3 What this result does and does not establish

120 The result shows that the rational architecture supports high closed-loop specialist capability. It is an  
121 existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization  
122 or state of the art.

# 123 4 Exact weight-derived structure through attention

## 124 4.1 Exact layerwise construction

125 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives  
126 a high-order coefficient table. Each entry says how strongly one combination of query, key, and  
127 value coordinates contributes to an output. It is *weight-derived* because no examples or fitted probe  
128 are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and  
129 an independent explicit polynomial agree to  $2.00 \times 10^{-15}$ . The internal core identity agrees to  
130  $1.24 \times 10^{-14}$ .

131 The deployed attention also normalizes each query and key with Equation 3. Each normalization  
132 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled  
133 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit  
134 numerator and denominator factors carry the normalization. The rational extension agrees to  $3.61 \times$   
135  $10^{-16}$ . We then reconstruct all four vision and eight joint attention modules in each of three  
136 trained checkpoints, covering 384 heads in total. The worst relative  $\ell_2$  error is  $2.56 \times 10^{-7}$  and the  
137 worst absolute difference is  $2.23 \times 10^{-5}$ . This residual comes from the deployed implementation’s  
138 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

139 We separately certify the three released convolutional checkpoint files associated with the historical  
140 capability aggregates in Section 3. Their vision stacks contain no attention, so we independently  
141 rebuild all eight joint modules and all twelve heads per module from the saved weights. Sixteen  
142 deterministically selected cache inputs per checkpoint yield 384 module cases and 4,608 head-input  
143 cases. Every output is finite and the worst relative  $\ell_2$  error is  $6.36 \times 10^{-16}$ , far below the frozen  $10^{-6}$   
144 gate. This same-checkpoint certificate covers the joint-attention equation at each normalized layer  
145 input. It does not rebuild the convolutional stem, feed-forward blocks, residual composition, final  
146 normalization, or action head as one end-to-end contraction.

147 Naively materializing the real width-384 coefficient tensor would require approximately  $3.2 \times 10^{15}$   
148 values. We instead contract the CP factors into an  $O(d^2)$  reduced Gram. This smaller matrix plays  
149 the role of a covariance matrix: its leading eigenvectors rank the directions that carry the most  
150 coefficient energy without constructing the full table. At toy multi-head scale, it matches brute-force  
151 materialization to  $2.13 \times 10^{-14}$ .

## 4.2 Trained policy result

For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all evaluation examples. The weights choose the subspace. Real cached activations are used only afterward to measure how faithfully that subspace preserves the layer’s computation.

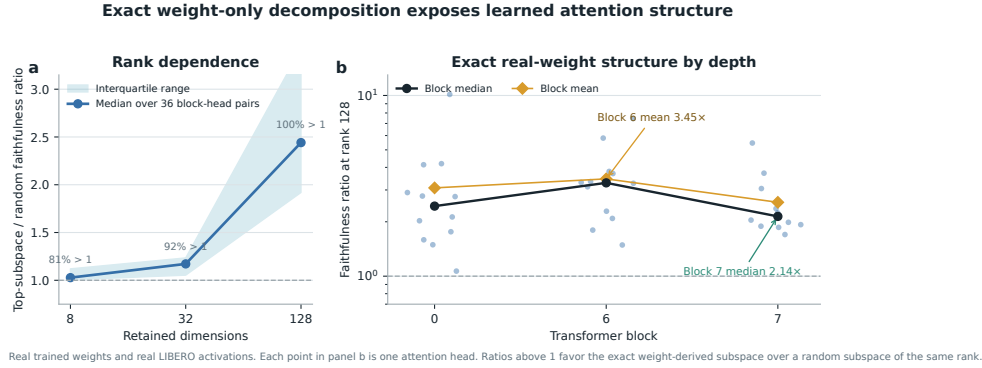


Figure 4: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

## 5 Related work

**VLA capability.** OpenVLA established an open pretrained VLA and standardized LIBERO comparisons [1]. OpenVLA-OFT showed that continuous action chunks and simple regression can substantially improve adaptation and control efficiency [2]. These systems use large pretrained backbones.  $\chi$ -VLA instead studies a compact from-scratch specialist under an algebraic restriction.

**Tensor and polynomial networks.** Polynomial networks and tensor networks use explicit multilinear structure for representation and learning [12–14]. Bilinear MLP work showed that weight eigendecomposition can expose low-rank features and circuits [3].  $\chi$ -nets introduced ODT for compositional bilinear networks [4]. We extend the setting to a continuous closed-loop policy and give an exact reduced construction for individual softmax-free attention layers.

**Causal mechanisms and policy analysis.** Bilinear IMPALA policies have combined competitive ProcGen control with weight-derived eigenfilters, followed by activation probing and control [5]. This is the closest architectural precedent for weight-based policy analysis. Recent work steers pretrained VLAs through causally identified activation directions [6]. Sparse feature circuits identify and edit causal feature graphs in conventional language models [7], while nonlinear causal reductions recover behavioral patterns in reinforcement-learning policies from rollout perturbations [8]. These results make a universal separation from conventional policies untenable. Our intended distinction is empirical: exact weight auditing in a strong continuous vision-language policy.

183 **VLA robustness.** Recent evaluations show that standard LIBERO success can coexist with weak  
 184 robustness and language use [9–11]. We therefore treat LIBERO-Object as a bounded specialist  
 185 capability setting rather than evidence of broad generalization.

## 186 6 Limitations and conclusion

187  $\chi$ -VLA demonstrates that a complete rational tensor policy can reach strong specialist closed-loop  
 188 capability. The real checkpoint passes a mechanical operator audit. Exact weight-derived analysis  
 189 now crosses individual attention layers.

190 The scope is deliberately specific. LIBERO-Object is one specialist suite with fixed task instructions,  
 191 so these results do not establish broad robotic generalization or robust instruction grounding. Exact  
 192 reconstruction is certified layer by layer and does not yet yield one compact symbolic contraction of  
 193 the full policy. The three-checkpoint ensemble trades three forward passes for its strongest capability  
 194 result.

195 Within that scope, polynomial and rational restrictions are compatible with strong closed-loop control  
 196 and provide a precise weight-derived analysis surface. The released convolutional files support  
 197 the joint-attention equation certificate and are associated with the historical capability aggregates.  
 198 Separate ViT checkpoints support the reduced-Gram subspace screen. We do not claim a direct  
 199 coefficient-edit interface.

200 **Reproducibility.** Training and evaluation use fixed task lists, explicit camera and action conventions,  
 201 canonical-state manifests, checkpointed moving-average weights, and per-run result files. Historical  
 202 Modal cache-backed analyses read dataset-native task metadata. The replacement Athena pipeline  
 203 pins that metadata and maps dataset identifiers to official identifiers before defining task splits. A  
 204 source-level audit compared all 271,996 cached frames across Object, Spatial, Goal, and LIBERO-10  
 205 with their pinned LeRobot revisions. Task identifiers, language joins, states, actions, and resized  
 206 images matched exactly for every frame. The final artifact will release code, checkpoint hashes, exact  
 207 attention certificates, and per-episode evaluation logs. Preflight checks fail if canonical initial states  
 208 cannot be loaded.

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## A Supporting results and decisions

### A.1 Per-task ensemble behavior

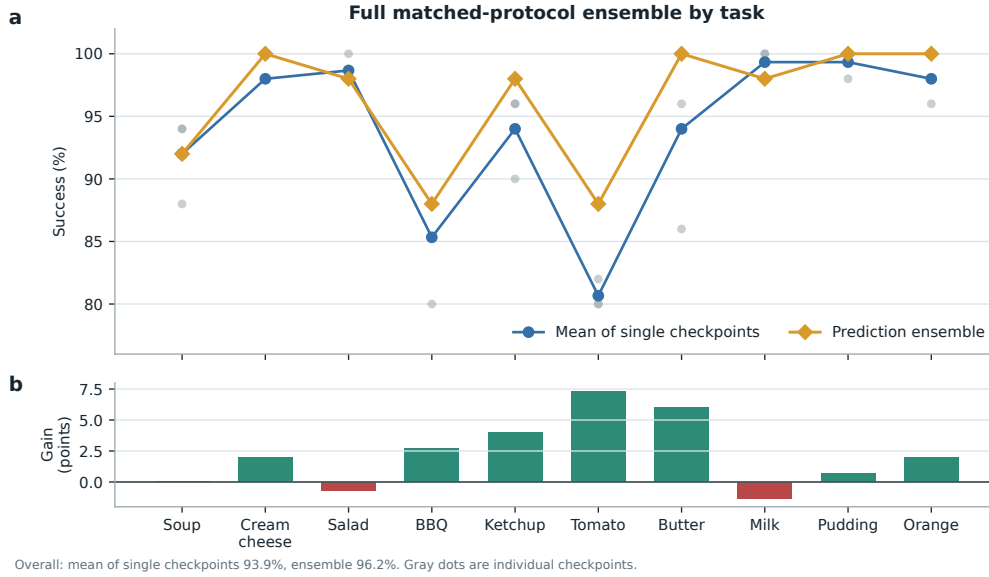


Figure 5: **Per-task effect of prediction averaging in the paired ensemble evaluation.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 19.2M-parameter policy.

### A.2 Expanded attention screen

The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.



### 242 A.3 Numerical scope of rational conversion

243 The deployed degree- $[2/2]$  least-squares rational fit is exact as a rational computation even though it  
 244 approximates RMS normalization. Its numerator has coefficients (5.32997, 7.53552, 0.327274) and  
 245 the denominator has coefficients (1, 9.64827, 2.60664). Every denominator coefficient is positive,  
 246 which certifies  $Q(v) > 0$  for all  $v \geq 0$ . We instrument all 53 RationalNorm module instances,  
 247 each invoked once per forward pass, on 4,096 deterministic inputs for each of three convolutional  
 248 checkpoints. Across 544,542,720 activation rows, the minimum observed denominator is 1.0016 and  
 249 no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass that  
 250 recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric, defined  
 251 as the maximum of normalized- and physical-action NRMSE, is  $2.414 \times 10^{-6}$  against a frozen  $10^{-3}$   
 252 gate.

253 The certificate also bounds what the operator approximates. Observed normalized mean-square ratios  
 254  $v$  span  $1.66 \times 10^{-4}$  to 17.17, outside the nominal fit interval  $[0.1, 10]$ . Across site-checkpoint pairs,  
 255 the minimum coverages are 60.018% in that interval and 60.022% with rational-versus-exact-RMS  
 256 scale error at most 3.4%. The frozen 99% secondary coverage gates therefore do not pass. The  
 257 positive claim is therefore global absence of nonnegative denominator poles and cache-conditioned  
 258 floating-point fidelity to the deployed rational operator, not uniform agreement with exact RMS  
 259 scaling. Matched alternative-normalizer runtime overhead remains unmeasured.

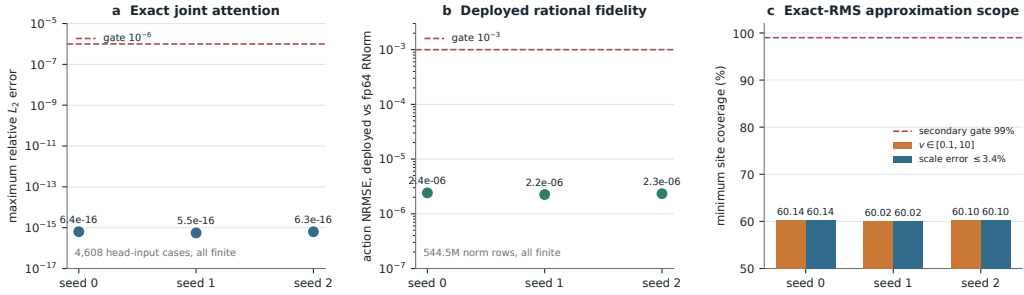


Figure 6: **Frozen structural certificates on the released convolutional files associated with the historical capability aggregates.** Panels (a) and (b) pass their primary gates with large margins. Panel (a) reconstructs all 4,608 audited joint-attention head-input cases. Panel (b) compares deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64. Panel (c) reports the separately frozen exact-RMS approximation coverage, which does not pass its 99% secondary gate.